Calculus I

Computation problems #12

Topics:

- Alternating Series
- Power Series
- Taylor and Maclaurin Series

Problem 1. Find the radius and interval of convergence. For what values of x does the series converge absolutely? conditionally?

$$\sum_{n=1}^{\infty}(-2)^n(n+1)(x-1)^n.$$

Problem 2. Find the radius of convergence for the series

$$\sum_{n=1}^{\infty}\frac{n!\,x^n}{3\cdot 6\cdot 9\cdots 3n}.$$

Problem 3. Find the Maclaurin series for the functions $f(x)=5\cos\pi x$.

Problem 4. Find the Taylor series generated by $f(x)=2^x$ at $x=1$.

Problem 5. Use the Taylor series for $\dfrac{1}{1-x^2}$ to obtain a series for $\dfrac{2x}{(1-x^2)^2}$.

Problem 6. Consider the series $\displaystyle\sum_{n=1}^{\infty}a_n$ defined by the formula

$$a_1=1,\ \ a_{n+1}=\frac{n(n+1)}{(n+2)(n+3)}a_n.$$

Is this series convergent? Justify your answer.

Problem 7. Find all values of x for which

$$\sum_{n=1}^{\infty}\frac{nx^n}{(n+1)(2x+1)^n}$$

converges absolutely.

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Answer to selected problems

1. Converges absolutely $\frac{1}{2} < x < \frac{3}{2}$, no x for converging conditionally
2. 3
3. $\sum_{n=0}^{\infty} f^{(n)}(0)\frac{x^n}{n!} = \sum_{k=0}^{\infty} \frac{5(-1)^k(\pi x)^{2k}}{(2k)!}$
4. $\sum_{n=0}^{\infty} \frac{2(\ln 2)^n(x-1)^n}{n!}$
6. Converges
7. $x \in (-\infty, -1) \cup \Big(-\frac{1}{3}, \infty\Big)$.